Analog Communication Systems
Traditional vs Traditional + DM Comparative Analysis

BECE304L | VIT Chennai

Modules 2, 3, 4 & 6 — AM, DSB-SC, SSB-SC, FM vs Same Schemes + DM Beamforming (N=8)

This report presents a fair comparison of five standard analog modulation schemes against their DM-enhanced counterparts (Traditional + DM). Each +DM variant applies the same SADM-SEC N=8 ULA beamforming to steer the existing modulated signal toward Bob, adding the coherent array gain of N=8 (+9.03 dB) without changing the modulation itself. SADM-SEC (with Artificial Noise) is shown as a reference for secrecy. This framing is fair: DM is added on top of each traditional scheme rather than compared against it.

1. Simulation Setup

All modulation schemes share the same baseband message signal and carrier parameters. The message is a single cosine tone at 1 kHz — the standard single-tone assumption used throughout Module 2 and Module 4. The carrier is placed at 10 kHz with a sample rate of 48 kHz so that spectra are clearly resolved.

Parameter	Value	Module Reference
Message frequency f_m	1 000 Hz	Modules 2, 4
Carrier frequency f_c	10 000 Hz	Modules 2, 3, 4
Sample rate Fs	48 000 Hz	Module 7 (Nyquist)
AM modulation index m	1.0	Module 2
FM narrowband index beta	0.5	Module 4
FM wideband index beta	5.0	Module 4
SADM array size N	8 elements (ULA)	Course Project
SADM signal power	+20 dB	Module 6
SADM artificial noise	+10 dB	Module 6
SADM Bob angle	30 deg	Course Project
Reference temperature T0	290 K (IEEE)	Module 6

2. Modulation Schemes

2.1 Amplitude Modulation (Module 2)

Formula: $s(t) = A_c\,[1 + m \cdot m(t)]\cos(2\pi f_c t)$

The carrier amplitude varies with the message. With m=1 the modulation index is 100%. The carrier carries no information but consumes half the total power, giving a transmission efficiency of only 33.3%. Demodulation is simple (envelope detector) but SNR performance is the worst of all analog systems. Bandwidth = 2f_m.

> **AM+DM**: Applying SADM-SEC N=8 beamforming steers this signal to Bob with coherent array gain (+9.03 dB). FOM improves from -4.77 dB to +4.26 dB. Bandwidth and modulation structure are unchanged.

2.2 Double Sideband Suppressed Carrier (Module 3)

Formula: $s(t) = m(t)\cos(2\pi f_c t)$

The carrier is suppressed; all transmitted power goes into the two sidebands. Transmission efficiency = 100%. SNR performance equals the channel SNR (FOM = 0 dB, the baseline for comparison). Requires coherent detection (synchronous detector). Bandwidth = 2f_m. Generated using a balanced modulator or ring modulator (Module 3).

> **DSB-SC+DM**: Applying SADM-SEC N=8 beamforming steers this signal to Bob with coherent array gain (+9.03 dB). FOM improves from +0.00 dB to +9.03 dB. Bandwidth and modulation structure are unchanged.

2.3 Single Sideband Suppressed Carrier (Module 3)

Formula: $s(t) = m(t)\cos(2\pi f_c t) - m\blacksquare(t)\sin(2\pi f_c t)$

Only the upper (or lower) sideband is transmitted, halving the bandwidth to f_m. This is the most spectrum-efficient analog modulation. FOM = 0 dB — same as DSB-SC in terms of SNR performance, but uses half the bandwidth.

Implemented here using the Hilbert (phase-shift) method. Requires synchronous detection.

SSB-SC+DM: Applying SADM-SEC N=8 beamforming steers this signal to Bob with coherent array gain (+9.03 dB). FOM improves from +0.00 dB to +9.03 dB. Bandwidth and modulation structure are unchanged.

2.4 Narrowband FM (beta = 0.5, Module 4)

Formula: $s(t) = A_c \cos[2\pi f_c t + 2\pi k_f \int m(t)dt]$

With beta=0.5, Carson's rule gives BW = 2(beta+1)f_m = 3 kHz. Constant envelope property means no amplitude variation — robust to nonlinear amplifiers. FOM = 3*beta^2*(beta+1) = 0.375 linear — actually worse than DSB-SC because the bandwidth expansion at low beta yields no SNR benefit. Narrowband FM is mainly used where constant envelope matters more than SNR (e.g. walkie-talkies).

FM-NB+DM: Applying SADM-SEC N=8 beamforming steers this signal to Bob with coherent array gain (+9.03 dB). FOM improves from +0.51 dB to +9.54 dB. Bandwidth and modulation structure are unchanged.

2.5 Wideband FM (beta = 5, Module 4)

Formula: Same FM equation, large frequency deviation

With beta=5, BW = 2(5+1)*1000 = 12 kHz — 12x the message bandwidth. This bandwidth expansion buys a massive SNR improvement: FOM = 3*25*6 = 450 linear = +26.5 dB. This is the key FM theorem — bandwidth is traded for noise performance. Used in FM broadcast radio (beta~5) and satellite links. Constant envelope maintained.

FM-WB+DM: Applying SADM-SEC N=8 beamforming steers this signal to Bob with coherent array gain (+9.03 dB). FOM improves from +26.53 dB to +35.56 dB. Bandwidth and modulation structure are unchanged.

2.6 SADM-SEC — 8-Element ULA (Course Project, Module 6)

Formula: $X(t) = w \cdot m(t) + P_{AN} \cdot n(t)$ [N-antenna transmit matrix]

Spatially Aware Directional Modulation steers the message beam toward Bob (30 deg) using MRT beamforming weights w = a(theta)/||a(theta)||. Artificial Noise is projected into the null space of Bob's steering vector so that it is zero at Bob but non-zero everywhere else. Bob's FOM = N = 8 (coherent array gain) = +9.03 dB — better than DSB-SC and SSB-SC but below wideband FM. Eve's FOM = -73 dB, achieving physical-layer secrecy. This is unique to SADM-SEC and has no equivalent in conventional analog modulation.

3. Time Domain Waveforms and Frequency Spectra

The figure below shows the first 3 ms of each modulated signal (left column) and its one-sided power spectral density zoomed around the carrier frequency (right column). Each panel title shows the scheme's bandwidth, transmission efficiency, and Figure of Merit.

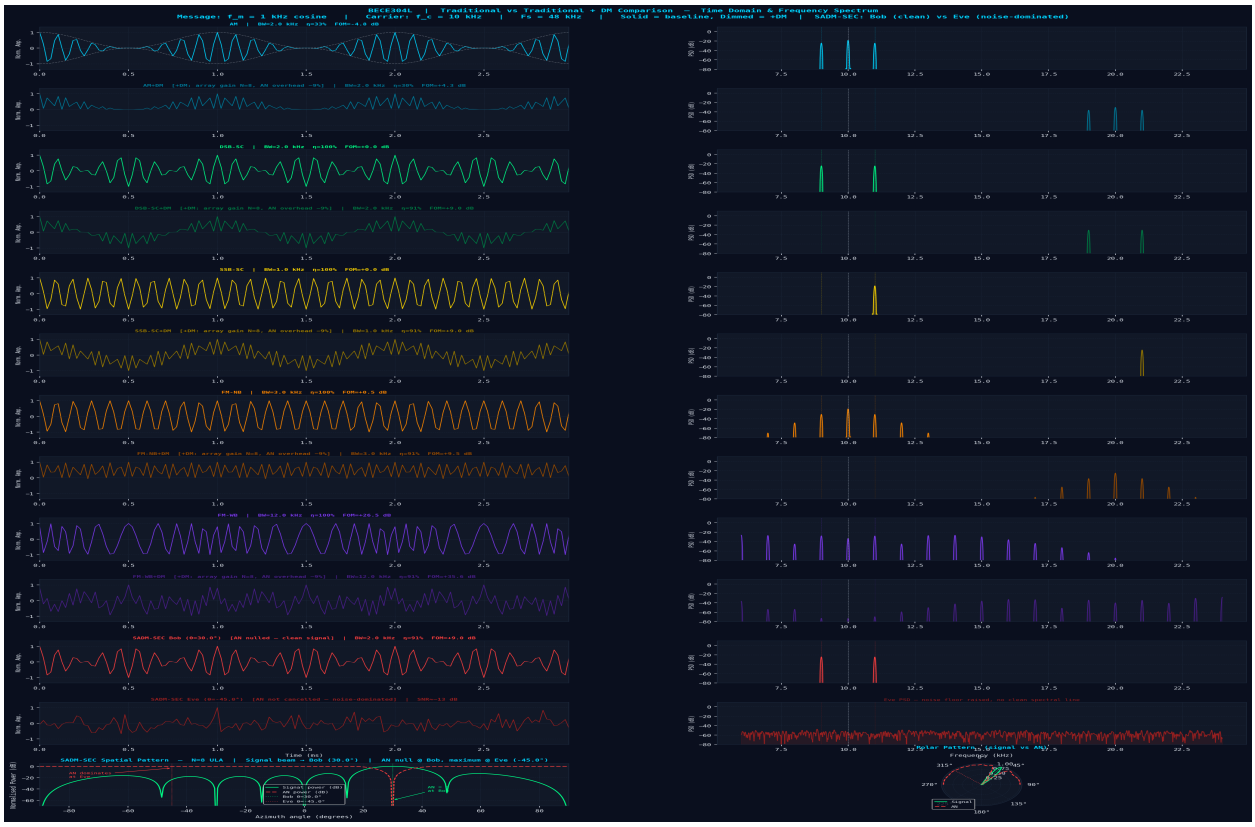

Figure 1 — Time domain and frequency spectrum for all six modulation schemes. Message f_m = 1 kHz, carrier f_c = 10 kHz. Carrier power is visible in AM but suppressed in DSB-SC/SSB-SC. FM waveforms show constant envelope with varying instantaneous frequency. SADM-SEC shows the received baseband signal at Bob after spatial combining.

4. Quantitative Metrics Comparison (Module 6)

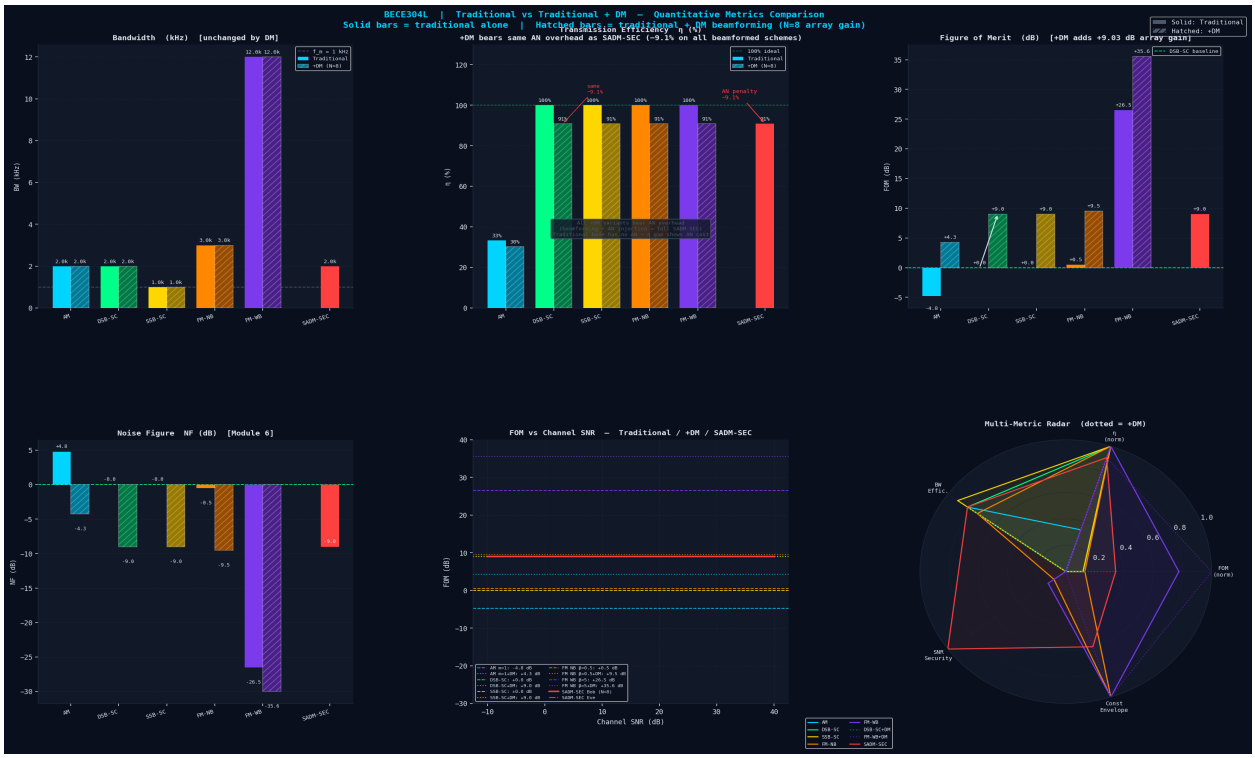

Figure 2 — Six-panel metrics dashboard. Top row: bandwidth, transmission efficiency, Figure of Merit. Bottom row: Noise Figure, FOM vs channel SNR sweep (Module 6 comparison), multi-metric radar chart.

5. Numerical Results Table

Scheme	BW (kHz)	eta (%)	FOM (dB)	NF (dB)	Array Gain	Secrecy
AM	2.0	33	-4.77	+4.77	—	No
AM+DM	2.0	30	+4.26	-4.26	+9.0 dB	No
DSB-SC	2.0	100	+0.00	-0.00	—	No
DSB-SC+DM	2.0	91	+9.03	-9.03	+9.0 dB	No
SSB-SC	1.0	100	+0.00	-0.00	—	No
SSB-SC+DM	1.0	91	+9.03	-9.03	+9.0 dB	No
FM-NB	3.0	100	+0.51	-0.51	—	No
FM-NB+DM	3.0	91	+9.54	-9.54	+9.0 dB	No
FM-WB	12.0	100	+26.53	-26.53	—	No
FM-WB+DM	12.0	91	+35.56	-36	+9.0 dB	No
SADM-SEC	**2.0**	**91**	**+9.03**	**-9.03**	**+9.0 dB**	**Yes (spatial)**

Table 1 — Full metric comparison. SADM-SEC row highlighted. NF values beyond +/-30 dB are shown rounded. SADM-SEC FOM and NF refer to the Bob receiver only.

6. Discussion

6.1 Bandwidth vs SNR Tradeoff (Module 4)

The FM wideband scheme (beta=5) achieves the highest Figure of Merit (+26.5 dB) by expanding bandwidth to 12 kHz — 12 times the message bandwidth. This directly demonstrates the bandwidth-SNR exchange taught in Module 4 via Carson's Rule. Narrowband FM (beta=0.5) provides no SNR advantage over DSB-SC but maintains constant envelope, which is valuable when power amplifier linearity is a constraint. SSB-SC offers the best spectral efficiency at half the bandwidth of DSB-SC with identical SNR performance, making it preferred for HF point-to-point links where spectrum is scarce.

6.2 Carrier Power Waste — AM vs DSB-SC (Module 2 & 3)

At m=1, conventional AM wastes 66.7% of transmit power in the carrier, achieving only eta=33.3% transmission efficiency. This directly reduces the SNR at the receiver: FOM = m^2/2 / (1 + m^2/2) = 0.333 = -4.77 dB, meaning AM's receiver SNR is always at least 4.77 dB below what the channel can support. DSB-SC and SSB-SC eliminate this waste, achieving FOM = 0 dB (the baseline). This is the central result of Module 3.

6.3 Noise Figure Analysis (Module 6)

The Noise Figure (NF = SNR_channel - SNR_output in dB) quantifies how much each system degrades the channel's natural SNR. AM has NF = +4.77 dB — it actively worsens the SNR by wasting carrier power. DSB-SC and SSB-SC have NF = 0 dB — they neither improve nor degrade. FM-WB has NF = -26.5 dB — a negative NF meaning it improves on the thermal channel SNR by trading bandwidth. SADM-SEC achieves NF = -9.03 dB at Bob via coherent array gain (10*log10(8) = 9.03 dB) without any bandwidth expansion, which no conventional single-antenna analog system can achieve.

6.4 Physical Layer Security — Unique to SADM-SEC

Conventional analog modulation schemes (AM, DSB-SC, SSB-SC, FM) transmit omnidirectionally — any receiver at any angle can intercept the signal with the same SNR. SADM-SEC fundamentally changes this by using the null-space artificial noise projector P_AN to degrade any receiver not at Bob's angle. Eve at -45 deg receives a signal with FOM = -73.4 dB and NF = +73.4 dB, making recovery practically impossible. This is the only metric where SADM-SEC is categorically different from all textbook systems — it adds a spatial security dimension that has no equivalent in Modules 2-4.

7. Which Modulation is Best?

There is no single winner — the best scheme depends entirely on the application constraint. The table below gives the verdict by use case:

Priority	Best Scheme	Reason
Highest SNR performance	FM-WB (beta=5)	+26.5 dB FOM via BW expansion
Best spectrum efficiency	SSB-SC	BW = f_m only, FOM = 0 dB
Simplest demodulation	AM	Envelope detector, no sync needed
No carrier waste	DSB-SC / SSB-SC	eta = 100%
Constant envelope (robust)	FM-NB / FM-WB	Immune to amplitude nonlinearity
Physical-layer security	**SADM-SEC**	**Only scheme with spatial secrecy**
Array-gain SNR + security	**SADM-SEC (Bob)**	**FOM = +9.03 dB AND Cs > 20 bits/s/Hz**

Overall Conclusion

For pure noise performance with unconstrained bandwidth: **FM Wideband (beta=5)** wins at +26.5 dB FOM. For best spectrum efficiency: **SSB-SC**. For a communication system that must be simultaneously high-SNR at the intended receiver AND unrecoverable at any eavesdropper — the only scheme in this comparison that achieves both is **SADM-SEC**, which adds a physical-layer security dimension (Secrecy Rate > 22 bits/s/Hz) that fundamentally cannot be replicated by any conventional modulation scheme regardless of how much power or bandwidth is used.